

D.I.Wireless Power and Signal Transfer

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Abstract

We wanted to build a circuit that uses magnetic fields to wirelessly transmit a signal, mimicking the telecoils in hearing aids. By modulating magnetic fields within two coils, we aimed to inductively transfer both power and signals through the air. We used concepts from our coursework in electromagnetism, solid state electronics, and communication systems to build this project. As a bonus, we set up our function generator for computer control and used this to create music.

Design Concept

We are using ring modulator mixers to transmit a message through magnetic fields. We use a function generator to create a 27kHz local oscillator frequency, and this LO is mixed with an audio message signal to create an amplitude modulated signal. This combined signal is used to vary the magnetic field in our transmitter coil. When the receiver coil is brought within close proximity of the transmitter coil, a magnetic field is induced within its windings which causes current to flow in the receiver coil. Sending the received signal through a ring modulator downconverts it back to the audio range. The downconverted signal then goes through a passive low pass filter and three amplifier stages before being sent into the speaker. We are using BJTs for power gain to drive the coils and op-amps to amplify the signals. All signals are generated using lab function generators. The transmitter and receiver coils are essentially an air-gapped transformer without a shared core.

Implementation

Step 1: We began by creating and simulating a theoretical design in LTSpice. The circuit was prototyped on a breadboard to allow for rapid troubleshooting of the oscillator and mixer stages.

Step 2: Using function generators and an oscilloscope, we tested our design and compared its performance against the simulation.

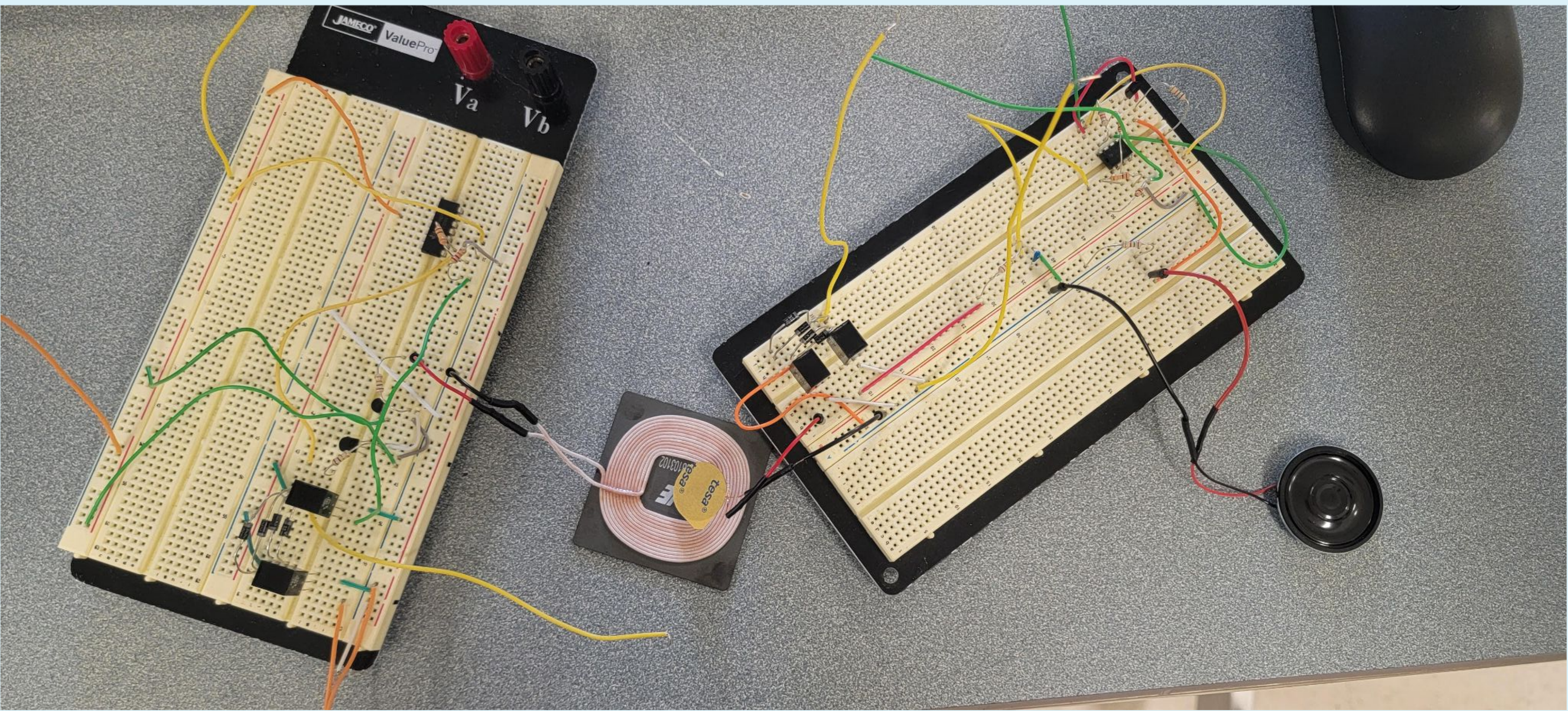
Technical Design

- Receiver and Transmitter coils
- 8 Schottky diodes
- 4 Center-Tapped Transformers
- BC337, BC327, 2N2222 Transistors
- LM358 and LM324 Operational Amplifiers
- Speaker (1W 8Ω)
- Resistors and Capacitors (assorted)
- Function Generators and DC Power Supply

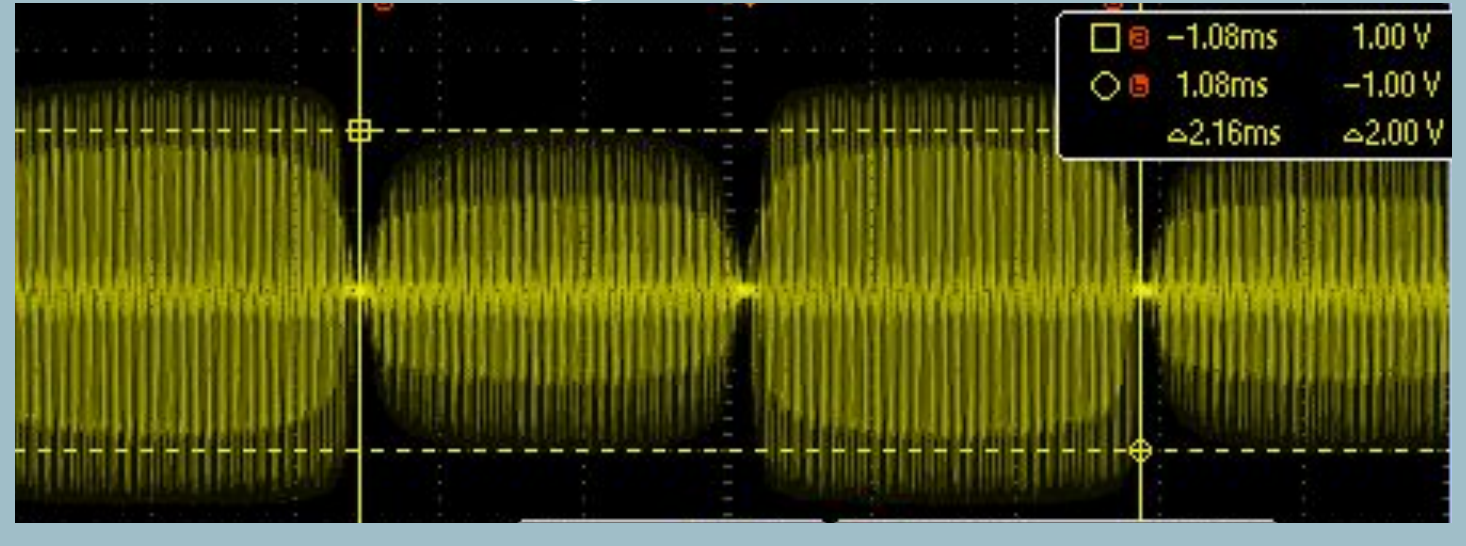
Testing and Validation

Step 3: During the testing phase, we realized that the signal was too weak to be picked up by the receiver. We decided to add an op-amp stage followed by a push/pull amplifier on the transmitter side to amplify the signal.

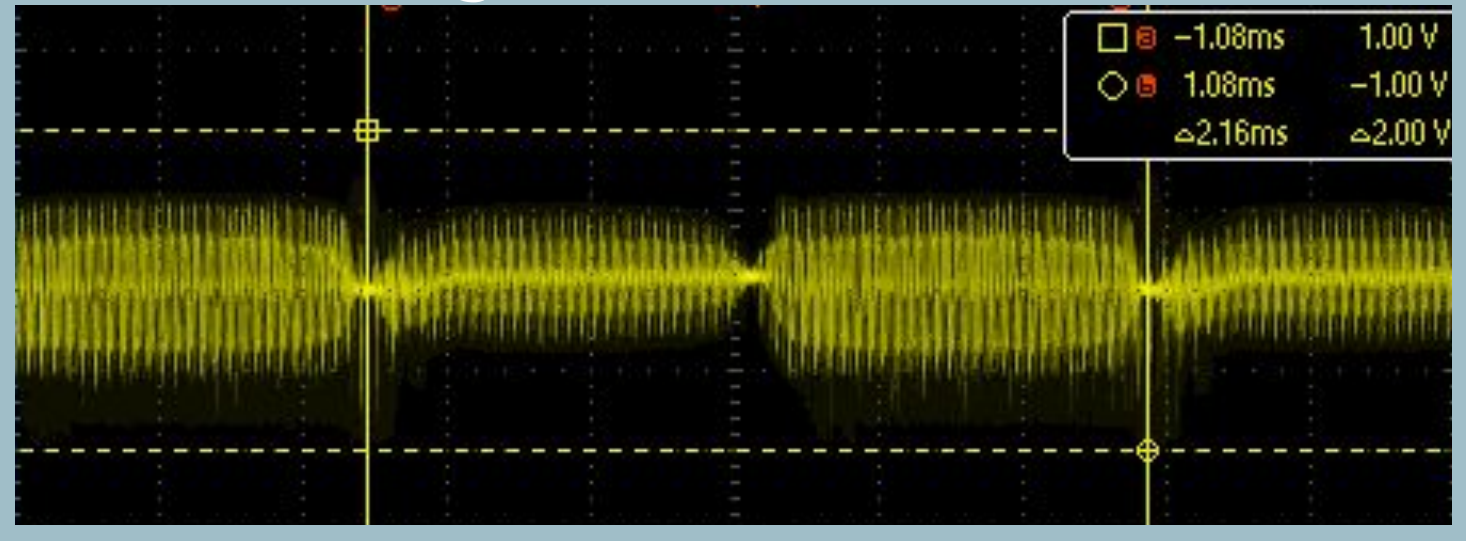
Step 4: Once the amplifiers were created, it was back to testing the signals to make sure they were high enough to drive a speaker, with 2 op-amp stages and an emitter follower right before the speaker. We experimented with a range of values to maximize our gain throughout the system.



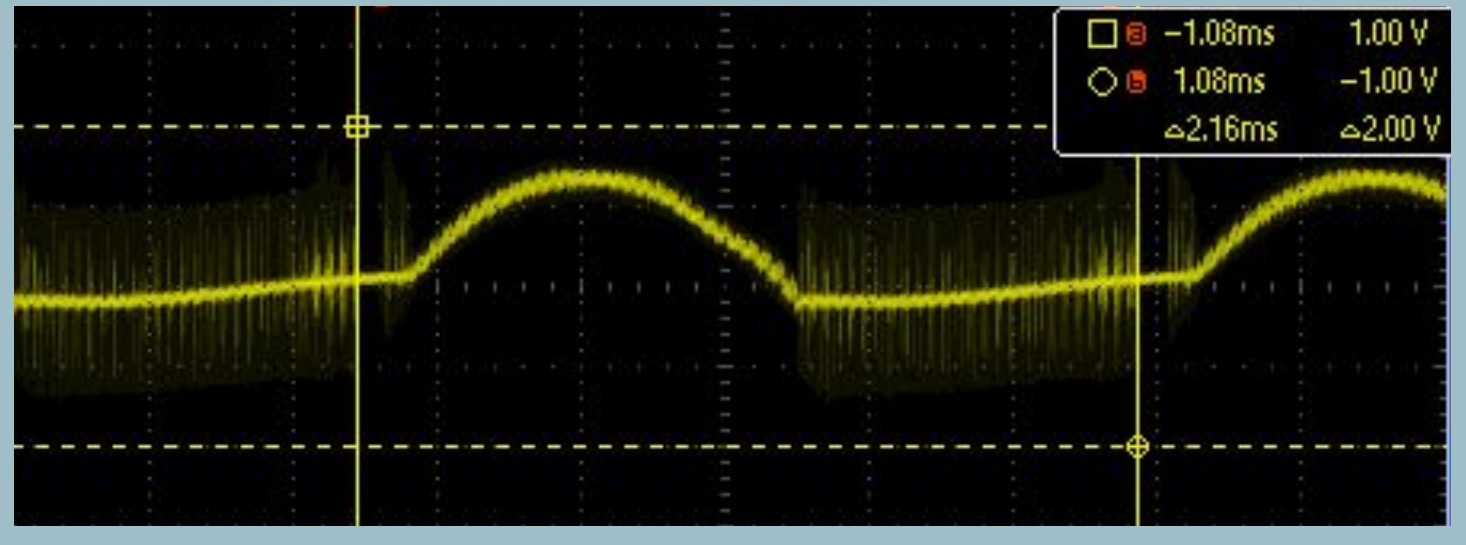
Transmitted Signal



Received Signal



Speaker Output



Conclusion

This project was a success! We were able to reach our goal of transmitting and receiving a signal using magnetic fields. We were unable to transmit as much power as we originally anticipated, but this wasn't an issue as our project became more signal oriented and cooler anyway.

<https://github.com/SamFOwens/D.I.Wireless>

References

[Understanding How Ring Modulators Produce AM Signals - Technical Articles](#)

[BJT Push-Pull explained - Electrical Engineering Stack Exchange](#)

Contribution

David: circuit build, jukebox program, debugging
Sam: schematic design, circuit build, debugging
Melissa: circuit build, piano program, kicad

